

Time Series Analysis: Session Learning Objectives

After this session, you will be able to

- 1 Recognise the main components of a time series
- 2 Analyse and explain given time series for these components
- 3 Employ graphical procedures to examine a time series for possible trends
- 4 Describe the trend in the series
 - Using a moving average
 - By fitting a line to the data using linear regression
- 5 Be aware of the upsides & downsides of finding trends by either method.

Examining Trends (1/2)

- 1 Assume time series data has at least 1 systematic pattern
 - Most common patterns are trends and seasonality.
 - **Trends:**
 - Trends are generally linear or quadratic.
 - Usual methods: moving averages or regression analysis
 - **Seasonality:**
 - A periodic fluctuation occurring at fixed & known intervals.
- 2 Unpredictable noise in real data obscures systematic patterns
 - Filters (e.g. smoothing) in time series analysis dampen errors.
 - However, Other patterns (e.g. residual effects of past observations or errors) can still remain.
- 3 Modellers have to account for all such patterns in their models.

Examining Trends (2/2)

- This week's lecture covers:
 - Identifying if there appears to be a trend
 - Describing and examining the trend using:
 - +> Moving Average Smoothing
 - +> Fitting a straight line to the data
 - A little bit on forecasting.
- Seasonal effects follow in detail in the next lecture.
- Cyclical effects are not covered as they can be difficult to identify and need long series

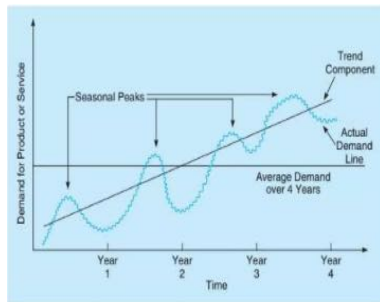


FIGURE 2.12: Demand Series with Several Components

Example 2.1: Trends

- Data from UK's Dept of Employment shows unemployed school leavers (in '000s)
- As 1979 began a recession, some patterns are apparent - but not from raw data.

Year	Jan to Mar	Apr to Jun	Jul to Sep	Oct to Dec
1979	22	12	110	31
1980	21	26	150	70
1981	50	36	146	110

TABLE 2.1: Unemployed school leavers in the UK

- In order to see these patterns clearly need to graph the data
- But first need to put data in list format

Example 2.1: Putting the Data in List Format

Year	Qtrs	Leavers
1979	1	22
1979	2	12
1979	3	110
1979	4	31
1980	1	21
1980	2	26
1980	3	150
1980	4	70
1981	1	50
1981	2	36
1981	3	146
1981	4	110

TABLE 2.2: Jobless School Leavers

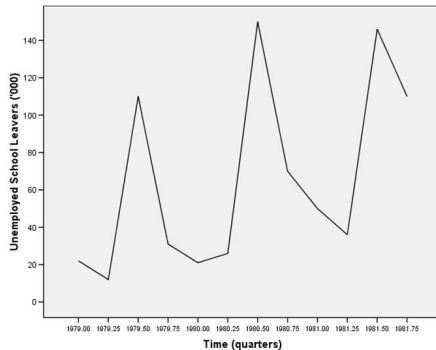


FIGURE 2.13: Jobless School Leavers

- Graph shows clear seasonality & trend effects

Estimating Trend I - Moving Averages

Ignore seasonality for now & concentrate on estimating trend

Moving averages (MA)

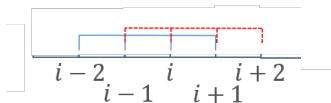
- Easily calculated and a powerful trend-spotting tool.
- In economics, a popular and often-used technical indicator.
- Can 'smooth' out local variation & possibly seasonal effects
- **Method:**
 - Replace each observation by a weighted average of neighbouring points
 - Results will depend on:
 - +> Number of observations used
 - +> Weights used
 - Certain subtleties about the method, however

Terminology of Moving Averages

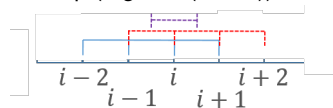
- Suppose we have the series: $x_1, x_2, x_3, \dots, x_n$
- Then the i th value of the 4 pt **Simple Moving Average (SMA)** is given by:

$$\bar{x}_i^{SMA} = (x_{i-2} + x_{i-1} + x_i + x_{i+1}) / 4$$

- This 4pt SMA is centred between points $i - 1, i$ (Fig 2.14(a, blue))
- Moving this SMA to the next points doesn't help (Fig 2.14(a, red))



(a) Red/Blue SMAs not on Datum i



(b) Purple SMA centred on Datum i

FIGURE 2.14: Simple Moving Average Windows

- The average of blue & red SMAs is centred on the datum point i (Fig 2.14(b, purple)).
- Better to align SMA and datum point.

Terminology of Moving Averages /2

- Should we always apply equal weight to each value?
 - Answer is not always clear cut
 - Might want to a greater weight on central points: 'Centred Moving Average'
 - You can verify that the i th value of the 'Centred Moving Average of order 4' (or 2×4 pt Moving Average) is:

$$\bar{x}_i^{CMA} = \frac{1}{8}x_{i-2} + \frac{1}{4}x_{i-1} + \frac{1}{4}x_i + \frac{1}{4}x_{i+1} + \frac{1}{8}x_{i+2}$$

- It is the purple moving average result from the previous page
- Moving Averaging is also known as a Sliding Window or Rolling Window Averaging
- Weights for moving averages must always add to 1.

Points to Note on Moving Averages

- For moving average, need window size (order), i.e. how many points?
- In general, for moving average of order r :
 - **Odd** r : MA values undefined for first $\frac{r-1}{2}$ & last $\frac{r-1}{2}$ series values.
 - **Even** r : MA values lie midway between observation points.
- For Quarterly Values natural to use order $r = 4$
- SMAs with $r = 4$ on quarterly data represent aggregated values:
 - These are reported at quarter end but refer to full quarter
 - They won't align on a data point (eg. Blue/Red SMAs in Fig 2.14)
- Better to have averages coinciding with observed data:
 - Otherwise, simple MA lags or leads the latest datum point by half the sample width.
 - This could have the effect of biasing the data in some way.

How Many Readings to Include in a Moving Average?

As seen, sometimes window contains a 'natural' number of readings (e.g. Quarterly Data), but, in general:

- No firm answer, but there is a trade-off to consider.
- Suppose the mean of the underlying process remains stable:
 - If window is too small, MA exhibits more variability than with larger number of data points.
 - In that sense, get more stability from including more points.
- If unexpected change occurs in the underlying process's mean:
 - With small window, MA tracks the changed process more closely than with more data points.
 - In that case, get more responsiveness by including fewer points.
- Ultimately a choice between stability and responsiveness.

Example 2.1 (Revisited): Calculating Moving Averages

Year	Quarters	Leavers	3-Pt Mean	Four-Pt MA	5-Pt WMA
1979	1	22	31/03 Value valid all Q1		
1979	2	12	48	43.75	
1979	3	110	51	43.5	43.63
1979	4	31	54	47	45.25
1980	1	21	26	57	52
1980	2	26	65.7	66.75	61.88
1980	3	150	82	74	70.38
1980	4	70	90	76.5	75.25
1981	1	50	52	75.5	76
1981	2	36	77.3	85.5	80.5
1981	3	146	97.3		
1981	4	110			

TABLE 2.3: Calculations on Table 2.2 data

- Verify a few of the values in the last 2 columns. . .

Example 2.1: Time Series & Fitted Trend

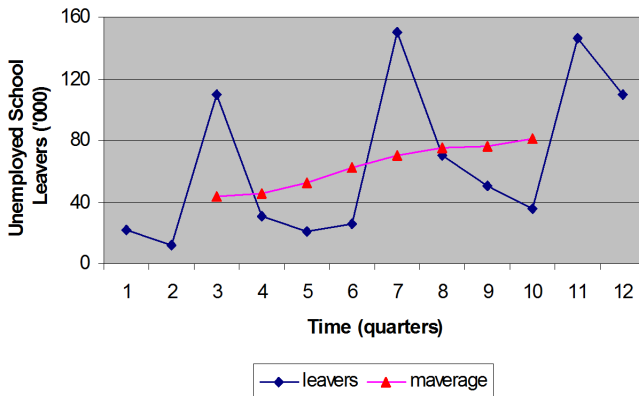


FIGURE 2.15: Quarterly School Leavers 1979-82 and Fitted Trend

- The increasing trend is now clearly visible from the moving average.
- We will see later centred moving averages used for forecasting.

Exponential Moving Average (EWMA)

- Probably most-used ad-hoc method to smooth time series
- **Advantages**
 - SMAs weight past values equally, EWMA assign exp'ly shrinking weights over time (blue vs red in Fig 2.16).

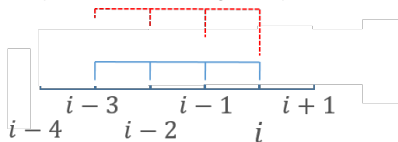


FIGURE 2.16: EWMA Vs SMA Weights Shown

- ‘Simple exponential smoothing’ (SES) is simplest such method.
- Approximates a simple ‘memory’ for recent & older values.
- Next smoothed value based on previous:
$$EMA_t = \alpha x_t + (1 - \alpha) EMA_{t-1} \text{ (} EMA_0 \text{ usually from SMA)}$$

EWMA (/2): A Bit of Math Underpinning

Expanding out EMA_t :

$$\begin{aligned}
 EMA_t &= \alpha x_t + (1 - \alpha) EMA_{t-1} \\
 &= \alpha x_t + (1 - \alpha) \times [\alpha x_{t-1} + (1 - \alpha) \times EMA_{t-2}] \\
 &= \alpha x_t + \alpha(1 - \alpha)x_{t-1} + \alpha(1 - \alpha)^2 x_{t-2} + (1 - \alpha)^3 EMA_{t-3}.
 \end{aligned}$$

- And so on until we reach the base term EMA_0 .
- Can show EMA_t is a weighted average of all previous observations
- Where the weight of observation $x_{t-r} = \alpha \times (1 - \alpha)^r$

α	$1 - \alpha$	$(1 - \alpha)^2$	$(1 - \alpha)^3$	$(1 - \alpha)^4$
0.9	0.1	0.01	0.001	0.0001
0.5	0.5	0.25	0.125	0.0625
0.1	0.9	0.81	0.729	0.6561

TABLE 2.4: Choosing Optimal α : Damping of Past Values

- Can see (Table 2.4) smaller α damp out past values slower & vice versa

EWMA (/3): Choosing The Smoothing Factor

k	1	2	3	4	5
α	1.0	0.67	0.5	0.4	0.33

TABLE 2.5: Smoothing Factor

$$\alpha = \frac{2}{k+1}$$

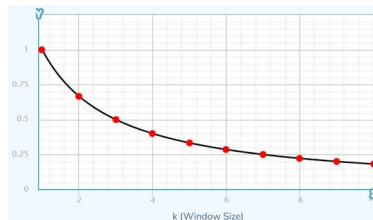


FIGURE 2.17: Visualizing window size effects

- Like other Moving Averages, EWMA has a window size (k) which dictates α
- Alternatively, α can be chosen.
 - +> Small $\alpha \approx 0$, past values get more weight.
 - +> Large $\alpha \approx 1$, recent observations are more important.

EWMA (/4): Various Window Sizes Compared



FIGURE 2.18: MSFT Data, EWMA With Varying Windows Jan '23-Dec '24

- Fig 2.18 shows EWMA of Microsoft Data using various Window Sizes.
 - EMA(20) (red) tracks the Closing prices best, not always a good thing.
 - Window sizes of 50 (blue), and 100 (violet) smooth the data better.
 - However, these lag data to a progressively greater extent.

EWMA (/5): Example 2.2 Saudi Oil Data

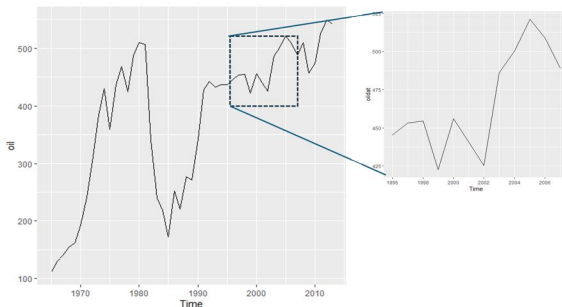
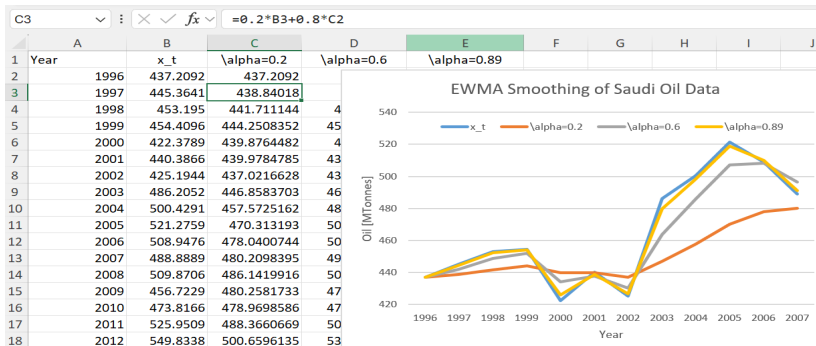


FIGURE 2.19: Saudi Oil Production Data (1960-2015)

- Note: EWMA/SES doesn't work well with trend, as here with full oil data
- In this case Double Exponential Smoothing must be used).
- No clear trend/ seasonality in Fig 2.19 extract ('96-'07) so SES is ok.
- R code can be found on loop

EWMA (/6): Example 2.2 Saudi Oil Data (Smoothed)



- Current EWMA based on present value & lagged EWMA $EWMA_{t-1}$
- Thus best to use a larger α if need to track the previous value
- Calculations given in loop in Excel file.

Moving Averages Compared



FIGURE 2.20: Microsoft Data, Moving Averages Aug '07-May '08

- Fig 2.20 shows EWMA(20), SMA(20), SMA(50) of MSFT Data.
- A Day-Trader should use **EWMA** due to its ability to track data changes.
- Investors with longer time horizons favour **SMA(50)** as it avoids 'false dawns' seen in **SMA(20)**.

Advantages & Disadvantages of Moving Averages

● Advantages

- Simple, flexible, fairly intuitive, easy-to-compute.
- Eliminates periodic data patterns when the moving average period matches the cycle (see Fig 2.15 (Jobless School Leavers))
- Ideal for determining seasonal, cyclic and irregular variations besides the trend values.

● Disadvantages

- No trend values for some time values.
- SMA not a math function so not very helpful in forecasting
- Tho EWMA has some applicability here (see below).
- Selecting optimal *order* of MA is also difficult.
- For non-linear trend values obtained can be biased in one/ other direction.

Estimating trend II - Fitting a Line

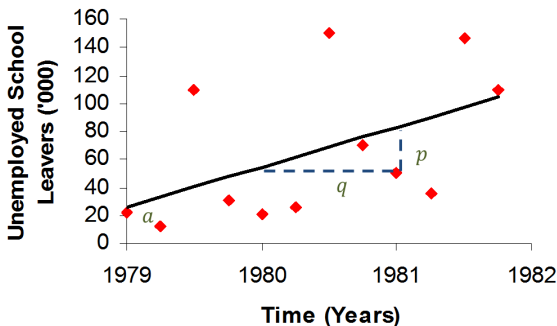
Another way to find the trend line:

- Find an equation for the 'best fit' to the trend (i.e. minimise errors btw value and line).
- The simplest is a straight line, i.e.

$$Y = a + bX$$

- a is the the Y axis intercept & b is the slope of the best fit line
- a, b minimize the sum of squared residuals.
- This is an example of so-called *Least Squares Method*
- Note:
 - Assume a linear trend in the data
 - Where this is not the case we have different formulas

Ex 2.1: School Leavers - Least Squares Regression



- Intercept a is shown on the graph.
- The slope b is given by $b = \frac{p}{q}$
- The equation of the line can be found to be

$$Y = 19.4 + 7.07X$$

- Note: Here, X is the time elapsed (in quarters) since 1979.

Forecasting - Simple Approach

- Return to 'best fit' line to the data, i.e. $Y = 19.4 + 7.07X$
- Future value of $X = X_i$, Q13, can find a value $Y = Y_i$

$$X_i = 19.4 + 7.07 \times 13 = 111.3$$

- However, many dangers here and care necessary:
 - Line fitting is part of a wider topic called 'Regression'
 - Assume repeat observations taken in time are **independent**. This is rarely the case.
 - Also assume prediction worsens as X used moves away from the known time values.
 - Shouldn't forecast without knowing of many statistical & other practical issues involved.

Pluses/ Minuses of (Linear) Least Squares Regression

● Advantages

- The method gives trend values for entire time period without omission.
- Trend line is a functional relationship between values & time
- This means future trend can be forecast.
- It is also a completely objective method.

● Disadvantages

- Calculations required make it tedious and complicated.
- Future forecasts by this method based only on trend values
- They ignore seasonal, cyclical or irregular variations.
- An extra point means a new equation has to be calculated.

Calculations for 'Best Fit' Line

- Let the equation of line be:

$$Y = b_0 + b_1 X \quad (2.1)$$

where Y_i are the to quarterly time values, incrementing by 1 Quarter.

- Then

$$b_1 = \frac{\sum_{i=1}^n (X_i - \bar{X}) (Y_i - \bar{Y})}{\sum_{i=1}^n \left[(X_i - \bar{X})^2 \right]}$$

- and

$$b_0 = \bar{Y} - b_1 \times \bar{X}$$

Lecture Summary

Given their importance and prevalence, we have focused on methods to find trends:

- Types of Moving Average Smoothing
 - Simple: Equal weights to all values in the window
 - Centred: Weigh outside values less compared to inner ones
 - Exponential: these give recent values greater (or lesser) weight
- 💡 Choice of Window size/ weights an issue for all Moving Averages
- Regression
 - Can be linear or non-linear
 - Future values of trend can be objectively calculated
- Both Moving Averages & Regression have their pluses/ minuses
- Each can estimate trends & forecast future values, albeit with caveats